

Real-World Aircraft Recognition Based on RF Fingerprinting With Few Labeled ADS-B Signals

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Abstract—Automatic Dependent Surveillance-Broadcast (ADS-B) is considered as a radar alternative in air transport systems, yet lack of authentication and encryption make it vulnerable to attack. In this paper, we propose a noise-robust radio frequency fingerprinting (RFF) approach for practical aircraft identification scenarios with a small sample size to address these issues. We develop a preprocessing method to improve noise-robustness by zero-padding useless signals and use a Siamese Networks-based few-shot training scheme for RFF recognition. The method is evaluated on a real-world ADS-B dataset, showing that signal preprocessing increases aircraft recognition accuracy by approximately 20% compared to using raw signals directly in small sample cases. Even with tens of labeled samples, our method achieves over 90% accuracy, outperforming other CNN identifiers by over 30%.

Index Terms—Radio Frequency Fingerprinting, ADS-B, Device Identification, Siamese Networks.

I. INTRODUCTION

Automatic Dependent Surveillance Broadcasting (ADS-B) system has gradually become the cornerstone of the next generation of digital skies. However, ADS-B lacks basic security mechanisms such as data encryption and message authentication which make it vulnerable to a large number of attacks such as spoofing, eavesdropping, interference, replay, and message modification [1]. Without authentication, interference or modification of navigation signals by attackers can compromise flight safety [2], especially false information injection. Despite the proposed encryption countermeasures for securing ADS-B, they conflict with its open nature and pose challenges in key management. By implementing robust authentication protocols, the ADS-B system can verify the identity of transmitting aircraft, prevent unauthorized access, and protect against data manipulation. Authentication enhances the overall security and reliability of air traffic communications, ensuring the integrity, confidentiality, and authenticity of the transmitted data [3].

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Thus, a broadcast-compatible authentication scheme is highly desirable for large-scale, dynamic ADS-B environments without the need for complex key management

Radio frequency fingerprinting identification (RFFI) uses intrinsic hardware features from a radio signal to identify the device where it comes from, providing a unique and difficult-to-forge identification method [4]. In recent years, deep neural network frameworks like convolutional neural network (CNN) [5] have shown great promise in automatically extracting features and achieving enhanced performance compared to traditional methods. However, these frameworks typically require a large amount of labeled data for effective training. When dealing with limited training samples, more complex hybrid neural networks become necessary. For instance, the deep reconstruction and classification network (DRCN) [6] leverages residual connections in CNN to facilitate training of deep networks. The generative adversarial network (GAN) [7] employs a game-theoretic setting, combining a generator and discriminator to generate realistic data samples. These methods prove valuable in scenarios with few training samples and contribute to improved performance. Although RFFI has been extensively studied for IoT devices, as of yet, only a few works have investigated its potential for aircraft identification.

Leonardi et al. classified seven types of aircraft using ADS-B message phase patterns but did not achieve individual identification [8]. Tu et al. created a real-world and large-scale dataset of ADS-B signals and used deep learning networks to achieve signal recognition [9]. Zha et al. proposed an RFF recognition method using contour stellar images, which achieved a classification accuracy rate of around 90% for five different aircraft [10]. However, the accuracy deteriorated rapidly at low SNR level. The negative effects of noise and interference on RFFI accuracy become more apparent with reduced labeled data and increased classification categories. Deep learning-based RFFI methods heavily rely on large datasets, making it vulnerable to overfitting with small datasets. Few-shot learning [11] can adapt models to new tasks with limited data, but few methods considered the application scenario of RFF identification for ADS-B signals.

Overall, practical few-shot RFF identification for aircraft remains an open challenge. To address this, we propose a novel RFF approach leveraging the distinctive waveform of ADS-B signals, to classify aircraft with a few labeled signals. The key contributions of this paper are:

- We propose a zero-padding (ZP)-based preprocessing algorithm to mitigate the effects of environmental noise and interference on ADS-B signals, leading to an easier and more efficient way for RFF feature extraction.
- We design a two-stage few-shot training scheme involving a Siamese Network for pre-training, an RFF extractor obtained from Siamese Network, and a Multi-layer Perception (MLP)-based feedforward network classifier trained on the RFF extractor. This training scheme is less susceptible to overfitting.
- Our proposed approach is verified on over-the-air ADS-B signals from aircraft in open and real-world scenarios, achieving over 90% accuracy with just tens of labeled samples. In small sample cases, the recognition accuracy of the signals using preprocessing increases by over 20% than using raw signals directly.

II. SYSTEM OVERVIEW

In this section, we overview the proposed RFF identification approach for the popular ADS-B data link standard. Raw I/Q signals are

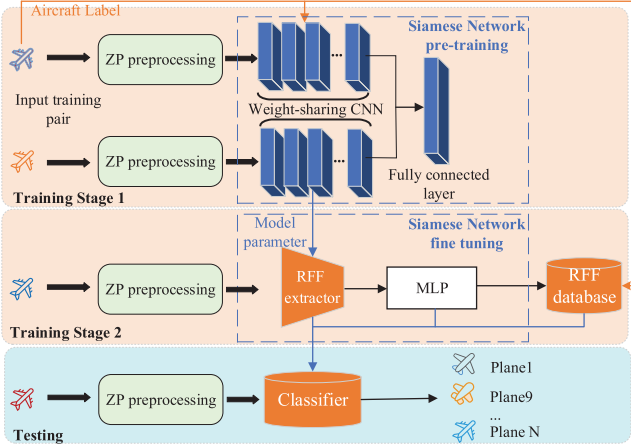


Fig. 1. Workflow of ADS-B signals classification.

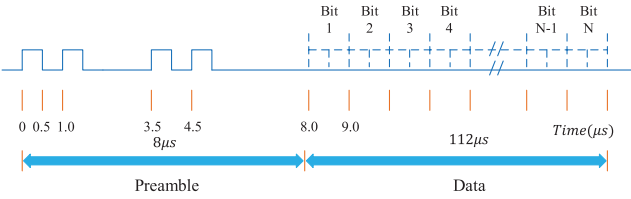


Fig. 2. Structure of ADS-B signal.

assumed to be obtained through existing signal detection algorithms [9]. Considering the successful experience of Siamese Networks and the scarcity of ADS-B signal classification applications, we attempt to use Siamese Networks for ADS-B signal classification. The system's workflow process is described in Fig. 1.

For both training stages, raw signals are first preprocessed to enhance the data quality. After that, the preprocessed I/Q signals are fed to Siamese Networks. Whereas a simple and effective training method for Siamese Networks, sampling from different pairs in the training set is applied to match the skewed distribution of similar and different pairs in the multi-class classification. Then the feedforward network classifier based on MLP is trained on the embeddings obtained from Siamese Networks instead of the nearest neighbor search, leading to an improvement in the prediction speed.

III. SIGNAL PREPROCESSING

The ADS-B considered in this paper is the 1090 MHz Extended Squitter (1090ES) mode which is a technology based on an S-mode transponder, with a frequency of 1090 MHz. The format of the ADS-B message data block utilizes Pulse Position Modulation (PPM) coding. The ADS-B signal consists of two parts, the 8μs preamble and the 112μs data block, as shown in Fig. 2. In the preamble, there are 8 microseconds of the signal header with a total of four fixed-position pulses, which can be used to detect and synchronize the ADS-B signal. The data block adopts the pulse position modulation method: a high pulse following a low pulse represents "1", and a low pulse following a high pulse indicates "0".

Real-world ADS-B signals suffer from environmental noise and interference, making it difficult to extract effective RF fingerprint features from raw signals. Since pulse modulation is used in ADS-B signals, the transmitting voltage has two levels. When the transmitting voltage is zero, the received signal is entirely made up of additive noise and interference. The lower level of the signal has harmful effects on feature

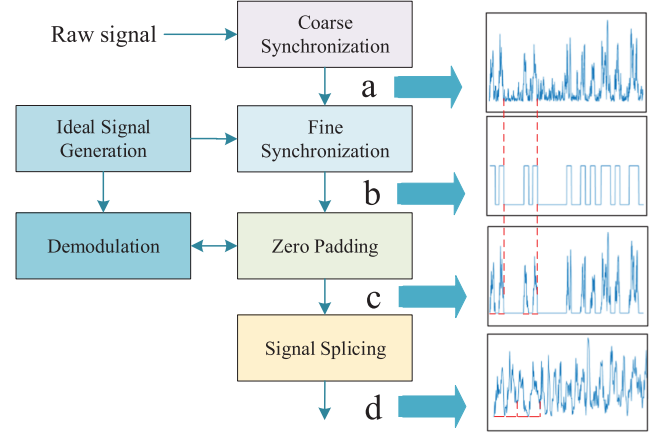


Fig. 3. Illustration of the preprocessing algorithm.

extraction. Inspired by this idea, we proposed a novel preprocessing algorithm, which improves the data quality by zero-padding these useless signals. Fig. 3 illustrates the block diagram and waveforms observed at each block of this preprocessing algorithm. To improve the effectiveness of this preprocessing algorithm, a coarse synchronization, as well as a fine synchronization, is utilized to detect the accurate region of interest before zero padding. The ZP preprocessing aims to extract informative signal segments and eliminate zero segments contaminated by additive noise and interference which consists of separate treatments for the preamble segment and the data segment.

First, the raw signal $s(i)$ is synchronized coarsely by detecting the fixed preamble. According to the modulation format of the ADS-B signal based on the 1090ES data link, the 8μs preamble contains four pulses, each with a width of 0.5μs. The four pulses are positioned at fixed intervals of 1.0μs, 3.5μs, and 4.5μs respectively. This enables us to directly determine the fixed pulse positions and derive the corresponding rectangular window waveform, denoted as $p(i)$, to align the preamble with the raw signal.

$$p(i) = \begin{cases} 1 & n f_s \leq i \leq n f_s + \frac{L}{2}, \\ 0 & \text{otherwise} \end{cases}, \quad (1)$$

where $n = 0, 1, 3.5, 4.5$ represents the arrival time of pulses, f_s is the sampling rate, L is the width of a pulse.

After synchronizing the preamble, a general positioning of the data segment is made. Based on the data segment modulation mode, a high pulse following a low pulse represents '1' and a low pulse following a high pulse indicates '0', we can demodulate the symbol bits of the data segment to initially locate the pulse position. We define the average energy of the first and second half of the n_{th} arriving data pulse ($n = 8, 9, 10, \dots, 120$) as $S_{n1} = \frac{2}{L} \sum_{i=n f_s}^{n f_s + \frac{L}{2}} s(i)$, where $s(i)$ is the raw signal. Since each half bit of the signal has length $\frac{1}{2}L$, $S_{n2} = \frac{2}{L} \sum_{i=n f_s + \frac{L}{2}}^{n f_s + L} s(i)$. Thus, we can obtain the signal frame $y(i)$ of the data segment to locate the signal by judging the symbol size of the first and second half signals in each bit symbol, S_{n1} and S_{n2} . If S_{n1} is greater than S_{n2} , the first half bit is assigned as "1" and the second half bit as "0". Conversely, if S_{n1} is less than S_{n2} , the bit assignments are reversed as shown in the following formulas.

$$y(i) = \begin{cases} 1 & n f_s \leq i \leq n f_s + \frac{L}{2}, S_{n1} > S_{n2}, \\ 0 & n f_s + \frac{L}{2} < i \leq n f_s + L, S_{n1} > S_{n2}, \end{cases} \quad (2)$$

$$y(i) = \begin{cases} 0 & n f_s \leq i \leq n f_s + \frac{L}{2} \\ 1 & n f_s + \frac{L}{2} < i \leq n f_s + L \end{cases}, S_{n1} \leq S_{n2}. \quad (3)$$

The actual arrival time of each pulse varies slightly due to many factors during transmission such as the channel state and the propagation distance. Therefore, we further use the data part to realize a fine synchronization. In actual communication scenarios, the arrival time of each pulse does not exceed 0.05us before and after. By comparing the average value in the first and second half of each symbol, the symbols in the data block will be demodulated.

$$\arg \max_v \frac{2}{L} \sum_{i=n f_s}^{n f_s + \frac{L}{2}} (s(i+v)), \quad (4)$$

where $v \in [-r, r]$ is the movement coefficient of the rectangular window, and r is the search range of the window. After fine synchronization, we can get the exact position of each pulse through a sliding correlation between the raw signal and the ideal pulse signal, leading to the derivation of the fine frame.

$$frame = \begin{cases} p(i) & v \leq i \leq 8f_s + v \\ y(i) & 8f_s + v \leq i \leq 120f_s + v \end{cases}. \quad (5)$$

The ideal pulse signal $frame$ with the demodulation information makes the rectangular window of the signal filter. According to (5), the overall length of $frame$ is $120f_s$, which is two-dimensional data just like the original signal. Then we compare the average values of the first and second half pulse and can define the half with a lower value as the useless part. By filtering the original signal with the rectangular window $frame$, part of the noise signal can be removed, leaving only the pulse signal containing fingerprints.

$$S_p = S_r \circ frame, \quad (6)$$

where $\circ$ represents the Hadamard product, $S_r = s_i(\bullet)^2 + s_q(\bullet)^2$ the original signal, $frame$ is the rectangular window, and S_p is the filtered signal.

Then, we splice the non-zero samples of the filtered signal to reconstruct the preprocessed signal S_n , that is, find all elements in S_p that are equal to 0 and remove them. Mathematically, it could be expressed as:

$$S_n = \{x \mid x \in S_p, x \neq 0\}. \quad (7)$$

By cutting off the noise part, we can concentrate the fingerprint information and increase the density of useful information, which is beneficial for the subsequent learning of networks.

After the signal is reconstructed and normalized, a slicing algorithm is exploited to realize data augmentation, which generates m slices from one frame of preprocessed samples. Denote M as the number of samples in one frame and L_w as the length of a slicing window, respectively. Each adjacent window has $L - \sigma$ overlapped samples. Then, the number of slices satisfies that $m = \lfloor \frac{M - L_w + 1}{\sigma} \rfloor$, where $\lfloor \cdot \rfloor$ denotes the floor operation. Each frame of the preprocessed signal is transformed into k overlapped slices through the sliding window, which largely extends signal characteristics. After slice processing, the preprocessed data S is finally obtained.

IV. CLASSIFICATION VIA SIAMESE NETWORKS

Siamese Networks are composed of two identical neural networks connected by an energy function, calculated using a metric between learned representations at the highest layer. Both networks share the same architecture and weights, acting as RFF extractors. Fig. 4 shows

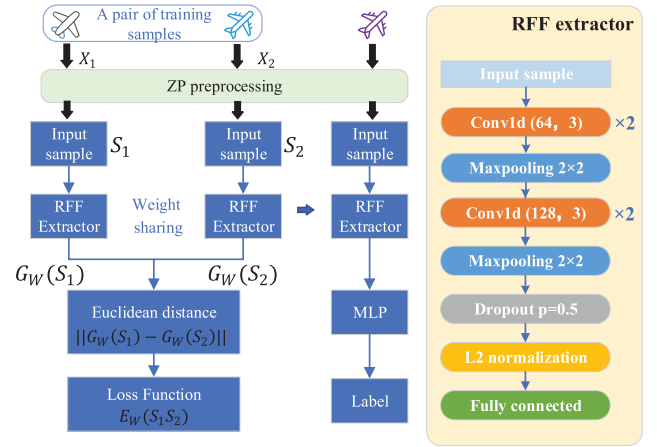


Fig. 4. Architecture of the classification model.

input sample pairs X_1 and X_2 from aircraft signals, with a binary label $Y \in (0, 1)$ indicating whether the pair is from the same class or not. The input pairs are preprocessed to obtain signals S_1 and S_2 , which are then fed into the networks to extract RFF features.

$G_w(S_1)$ and $G_w(S_2)$ are the outputs of the inception RFF extractors, whose responsibility is transforming the original sample to the characteristic space. Siamese Networks' model parameters W are shared by both RFF extractors. This procedure reduces the number of parameters to learn, resulting in less overfitting and requiring less data to train. Siamese Networks compare the similarity of data pairs and slices in data preprocessing which significantly increases the amount of data. Therefore, features can be efficiently extracted without a complex network architecture, reducing computational complexity. Our proposed inception network for feature extraction and processing is inspired by the structure of VGG16 Net, consisting of four convolutional layers, two max-pooling layers, and one dropout layer. The first convolutional layer uses 64 filters, and the second layer uses 128 filters. Both convolutional layers are activated by rectified linear unit (ReLU) and padding is used. After the convolutional layers, a max pooling layer and a dropout layer are following.

To measure the correlation between S_1 and S_2 , an energy function called $E_w(S_1, S_2)$ is used to calculate the low-dimensional feature similarity between the two samples. It uses the Euclidean distance of the mapping function as the similarity measure.

$$E_w(S_1, S_2) = \|G_w(S_1) - G_w(S_2)\|. \quad (8)$$

The loss function of the network is designed to minimize $E_w(S_1, S_2)$ when S_1 and S_2 are from the same class, and to maximize it when they are from different classes. By using the Euclidean distance as a similarity measurement function, the loss function can be rewritten as follows, where the true label Y (1 for similarity and 0 for dissimilarity), $(Y, S_1, S_2)^i$ denote the i th sample pair, the hyperparameter m represents the constraint.

$$L(W, (Y, S_1, S_2)^i) = \frac{1}{2}(1 - Y)(E_w(S_1, S_2)^i) + \frac{1}{2}Y \max(0, m - E_w(S_1, S_2)^i)^2. \quad (9)$$

After Siamese Networks training, the resulting output label is 0 or 1, which should be further classified into accurate tags. In metric learning, classification methods such as Cosine similarity, K-nearest neighbor (KNN), and multilayer perceptron (MLP) are commonly employed. Cosine similarity quantifies the similarity between vectors in metric

learning systems, as demonstrated by Seyed et al. in their approach that measures feature embeddings' similarity extracted from images or video frames [12]. KNN offers simplicity and interpretability, classifying samples based on majority labels from nearest neighbors, effectively handling complex decision boundaries and diverse data distributions. Shen et al. integrated KNN into a LoRa RFF recognition system [13]. However, the KNN approach can be time-consuming and increase memory usage for embeddings. In contrast, MLP classifiers excel at learning complex nonlinear decision boundaries, capturing intricate input-output relationships in high-dimensional data. In our two-stage approach, a two-layer MLP is trained on RFFs from Siamese Networks, fine-tuning its parameters through two ReLU-activated hidden layers and a softmax-based output layer for accurate predictions. The adoption of MLP reduces memory overhead, enabling faster inference during the classification phase.

In the first training stage, the Siamese Networks are trained by conducting classification based on a pair of inputs, which effectively augments the training data and mitigates the issue of model overfitting. In addition, weight decay and L2 regularization are used in the network model to reduce overfitting. In the second training stage, the RFF extractor with a good initialization is combined with a two-layer MLP. This fine-tuning process facilitates the precise classification of test samples, leading to subsequent improvement in classification accuracy and less overfitting compared to direct Siamese Networks.

V. EXPERIMENTS AND DISCUSSION

In the experiment, we utilized a real-world aircraft dataset of ADS-B signals from [9], which was collected at the China Civil Aviation Science and Technology Industrialization Base from June 3, 2020, to June 23, 2020. The raw signals consisted of a total of 13,000,000 samples distributed among 198 categories representing different aircraft. Each sample was recorded at a sample rate of 50 Msps and had a length of 3,000, comprising two-dimensional IQ data, with aircraft IDs serving as labels. The raw data underwent preprocessing, including resampling, header search, information decoding, CRC check, positioning, and labeling, to generate a comprehensive dataset of ADS-B signals. To create small sample datasets for evaluation, we randomly selected samples from the large dataset and formed sets with 10 categories. Within each category, we constructed datasets containing 10, 20, 50, 100, 150, and 300 samples per class. Additionally, we used 1/5 of the training sample size to create the test set.

We compared our proposed method with CNN [5], DRCN [6], and QGAN [7], while the effect of zero-padding preprocessing is also verified. To ensure fairness, we used the same training parameters and basic network architecture. Fig. 5 shows the identification accuracy of various methods for a real-world ADS-B dataset with varying numbers of labeled samples. The results indicate that ZP preprocessing improves performance, particularly when labeled samples are limited. As the number of samples used for training decreases, the effect of ZP preprocessing on accuracy improvement becomes more significant. With only 10 samples per device, preprocessing can improve performance by more than 20%. Because zero-padding effectively reduces noise by eliminating useless components and preserving important signal characteristics. By enhancing the signal quality, it improves the accuracy of feature extraction and classification. While CNN has the lowest identification accuracy due to limited data distribution information, semi-supervised methods such as DRCN, and QGAN improve identification performance by approximately 3% compared to CNN on small sample sizes. However, our proposed Siamese Networks outperform all other methods in any case of sample size, especially

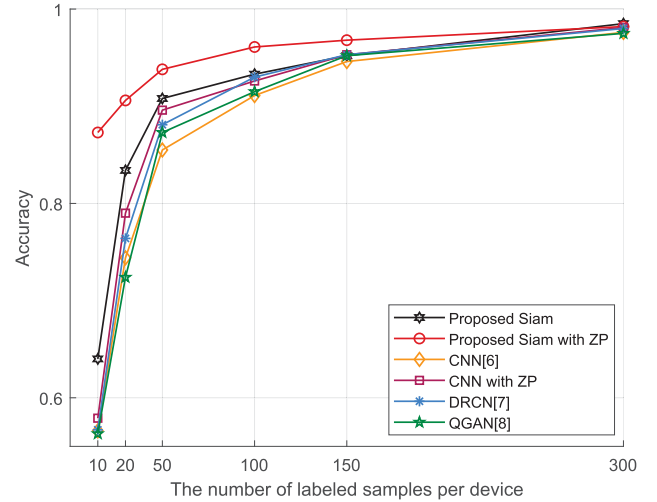


Fig. 5. Test accuracy comparison with different numbers of the labeled samples per device under different models.

TABLE I
CLASSIFICATION PERFORMANCE USING DIFFERENT PREPROCESSING METHODS

Preprocessing method	Accuracy
Zero-padding	88.3%
Wavelet transform [14]	65.3%
Savitzky-golay filter [15]	68.6%
PCA [16]	47.8%
FastICA [17]	56.7%

on small sample size datasets. Even with only 10 labeled samples, our proposed method achieves an accuracy of approximately 90%. In contrast, conventional methods experience significant degradation in identification performance.

We also conducted a comprehensive comparison of our proposed method with other commonly used noise reduction techniques applied to signals, including Wavelet Transform [14], Principal Component Analysis (PCA) [16], Savitzky-Golay(SG) filtering [15], and FastICA [17]. The signals were preprocessed using each of these methods, and the same network structure and parameters were used for training. The classification accuracy of each method was evaluated under the standard of 10 samples for each class, and the results are summarized in Table I. Our proposed method demonstrated superior performance for ADS-B signals compared to the other noise reduction techniques, showcasing its effectiveness in enhancing the classification accuracy and robustness of the system. The application of other noise reduction methods, such as Wavelet Transform, SG filtering, PCA, and FastICA, may not be suitable for ADS-B signals. Wavelet Transform's effectiveness depends on the choice of wavelet functions and its sensitivity to noise variability leading to suboptimal denoising. SG filtering provides limited improvement through signal smoothing, but its impact is constrained. PCA assumes linear data distribution, which may not be appropriate for complex nonlinear relationships in ADS-B signals, and it is sensitive to outliers. FastICA handles nonlinear relationships but is sensitive to parameter initialization. Moreover, signal separation methods like PCA and FastICA lack interpretability and may remove RF features from the signal. In contrast, the proposed zero-padding preprocessing method is more suitable for enhancing noise-robust performance in ADS-B signal processing, effectively preserving RF fingerprint information for improved classification accuracy.

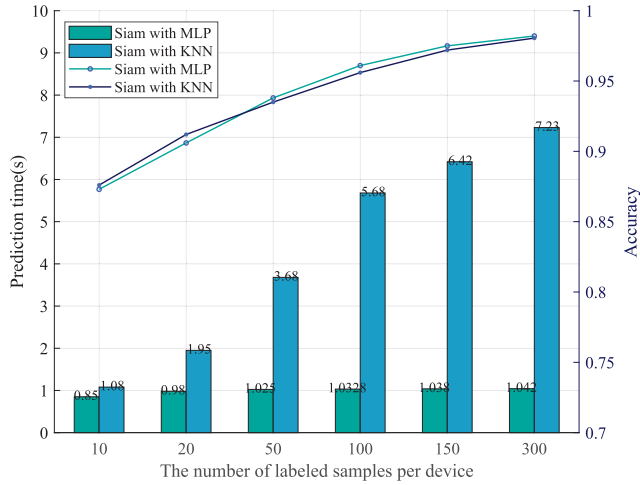


Fig. 6. Test accuracy and prediction time comparison with different numbers of the labeled samples per device with MLP or KNN.

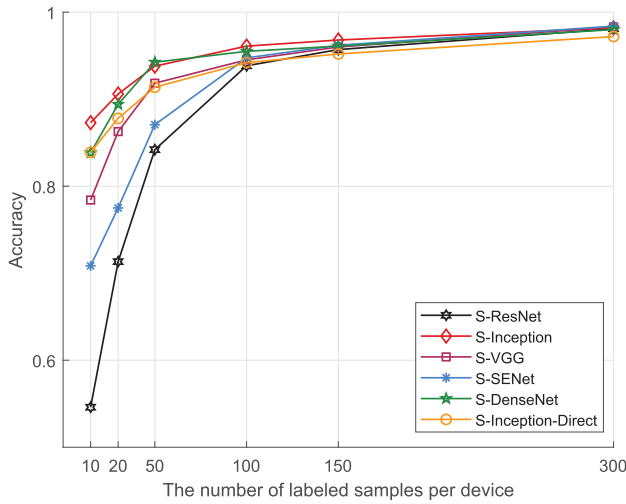


Fig. 7. Test accuracy comparison with different numbers of the labeled samples per device with inception models.

Fig. 6 displays the comparison of MLP and KNN combined with Siamese Networks in terms of identification accuracy and prediction time on a real-world ADS-B dataset with varying numbers of labeled samples. For testing purposes, we used 1/5 of the training sample size and ensured that the training model and parameters were uniform across both methods. The MLP method is faster in predicting even with an increase in the number of test samples, while the KNN method experiences a significant increase in prediction time due to the need to calculate the distance between each sample and the standard training results. Our study suggests that the combination of Siamese Networks followed by an MLP-based two-layer feedforward network is seven times faster than Siamese Networks with KNN, which is important for practical applications requiring real-time predictions. The test accuracy between the two methods is similar, and the classification performance primarily depends on the training model and parameters. Therefore, we recommend using the MLP and Siamese Networks combination for optimal classification performance.

In Fig. 7, we compared the performance of Siamese Networks with other CNN frameworks, including VGG, ResNet, SENet, and

DenseNet. To ensure a fair comparison, input samples of the same size were used, and the networks were fine-tuned. Compared with other CNN frameworks, the adjusted Siamese + Inception (S-Inception) network can provide competitive results, and particularly, the S-Inception combination achieves high accuracy in low-data scenarios. In scenarios with small sample sizes, the two-stage Siamese network training demonstrates superior performance compared to direct Siamese network training, achieving approximately 3% higher accuracy. This improvement can be attributed to better feature extraction and network fine-tuning, contributing to the enhanced performance. However, as the number of labeled samples increases, the performance gap between the two-stage and direct Siamese network training diminishes. In such cases, the direct Siamese network can also achieve satisfactory results by directly learning the embedding space and similarity metric from the data. Nevertheless, the two-stage training model retains advantages in terms of robustness and generalization due to its fine-tuning process. Overall, the two-stage training approach proves particularly beneficial in scenarios with limited labeled samples, offering better performance gains over direct Siamese network training.

To optimize the performance of Siamese Networks, it is recommended to utilize a simple network structure and employ a sufficient sampling pairing method when the sample size is small. This ensures that the Siamese Networks can effectively learn the differences between sample pairs, resulting in a better pre-trained model that is less prone to overfitting during secondary training. Conversely, in scenarios with sufficient samples, where a significant amount of information is available to the model during training, a more complex network can be employed, and sample pairs can be obtained through random sampling, accelerating the training process and improving classification accuracy. The abundance of labeled samples allows the model to learn complex patterns and relationships between input features and output labels, leading to better generalization and accuracy. Fig. 7 shows that for sample sizes of each class in the training set exceeding 100, the deeper network structure achieves slightly higher classification accuracy compared to the simpler network structure, requiring approximately one-third fewer training epochs, resulting in higher accuracy and faster fine-tuning.

VI. CONCLUSION

This paper addressed aircraft identification using RFF in practical scenarios with limited training data. A preprocessing algorithm based on zero padding enhances data quality for feature extraction. A two-stage training scheme was proposed with Siamese Networks initializing the RFF extractor, followed by MLP-based classifier training and fine-tuning. Experimental results on over-the-air ADS-B signals revealed significant performance improvements with the ZP preprocessing method and Siamese Networks, especially with small labeled sample sizes. However, the proposed method is only implemented on closed set real data. Future research will focus on applying few-shot learning in open sets to detect attacks.

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